



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2018 – 0134



Agricultural Machinery – Colloid Mill DYNAMICS JM80

Test Report Number	2018 – 0134
Date of Issuance	June 13, 2019
Valid Until	June 13, 2024
Agricultural Machine	Colloid Mill
Type	Vertical, Conical Burr Grinder
Brand	DYNAMICS
Model	JM80
Test Requested by	DYNAMICS DEVELOPMENT TRADE & GENERAL SERVICES INC. 312 Rivera St., Sta. Quiteria, Caloocan City

SUMMARY

Performance Criteria	AMTEC Test
Milling Capacity, kg/h	45.2
Noise Level, dB(A), Maximum*	83.3

* Allowable noise level for six (6) hours of continuous exposure based on Occupational Safety Health Standards, Department of Labor and Employment, Philippines, 2013.

OBSERVATIONS

1. Prime Mover	4 kW WENLINGSURIN Electric Motor
2. Operator's Manual	None
3. Previous Valid Test Reports of Similar Brand and Model	None
4. Transporting the Machine	The machine was transported manually by lifting and then displacing.
5. Adjustments	Not observed
6. Loading Input and Collecting Output	The hopper and output chute are accessible to the operators.
7. Number of Operators During the Test	1 Feeding Operator, 1 Manual Stirrer, 1 Bagger
8. Failure	No machine failure occurred during the test.
9. Safety	None
10. Other Remarks	None

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	May 16, 2018
Location of Test	AMTEC Building, Pili Drive, University of the Philippines Los Baños
Items Tested	Milling Capacity, Noise Level, and Power Consumption

METHODS OF TEST

The machine was tested based on AMTEC Methods of Test.

DESCRIPTION OF THE MACHINE

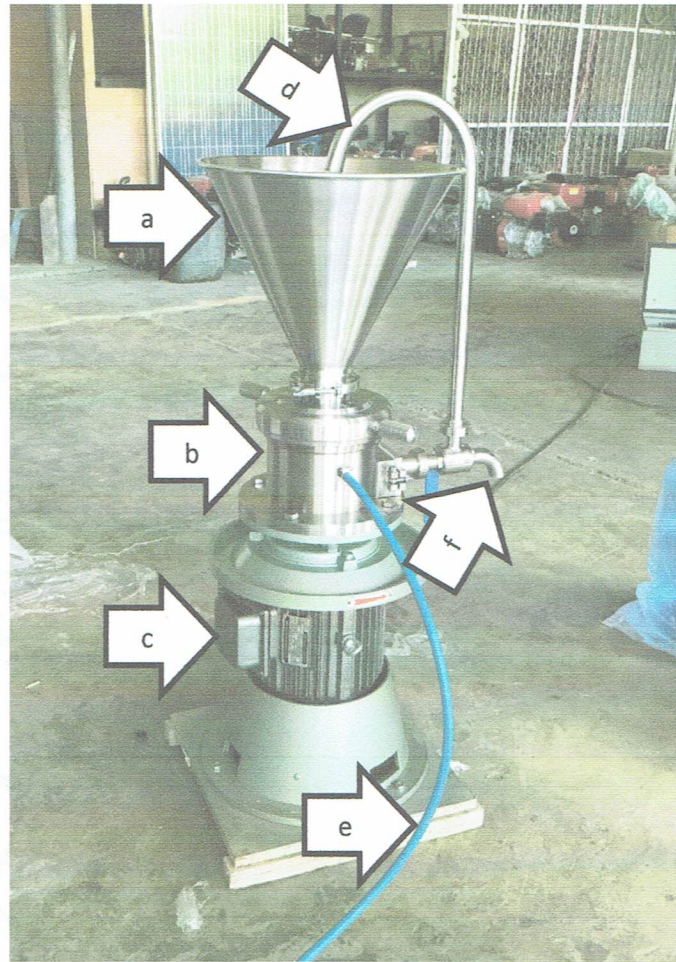


Figure 1. Parts of the DYNAMICS JM80 Colloid Mill:

- (a) Input Hopper
- (b) Milling Chamber
- (c) Prime Mover
- (d) Return Pipe
- (e) Cooling Water Supply Line
- (f) Cocoa Liquor Outlet

SPECIFICATIONS

Item	Manufacturer's Specifications ^a	AMTEC Verification ^b
1 Main Structure		
1.1 Overall dimensions, mm		
1.1.1 Length	ND	490
1.1.2 Width	ND	340
1.1.3 Height	ND	1070
1.2 Weight, without prime mover, kg	ND	NM
2 Rated Milling Capacity, kg/h	ND	45.2
3 Prime Mover		
3.1 Electric motor		
3.1.1 Brand	WENLINGSURIN	WENLINGSURIN
3.1.2 Model	Y112M-2	Y112M-2
3.1.3 Serial number	ND	ND
3.1.4 Make	ND	ND
3.1.5 Rated power, kW	4	1.78
3.1.6 Rated speed, rpm	3360	3595
3.1.7 Phase	3	3
3.1.8 Voltage, V	220	217.5
3.1.9 Current, A	14.35	10
3.1.10 Frequency, Hz	60	NM
4 Hopper		
4.1 Inlet Diameter, mm	ND	70
4.2 Height from the ground, mm	ND	980
4.3 Material	ND	Stainless Steel
4.4 Location	ND	Above Milling Chamber
4.5 Means of attachment	ND	Bolted
5 Recirculating Pipe		
5.1 Diameter, mm	ND	22
5.2 Height from the ground, mm	ND	962
5.3 Material	ND	Stainless Steel
5.4 Location	ND	Above Output Pipe
5.5 Means of attachment	ND	Threaded
6 Output Pipe		
6.1 Diameter, mm	ND	22
6.2 Height from the ground, mm	ND	500
6.3 Material	ND	Stainless Steel
6.4 Location	ND	Beside Milling Chamber
6.5 Means of attachment	ND	Bolted

Note: ND - No Data; NM - Not Measured

^a Data obtained from component nameplates^b Data obtained from actual test results and on-site measurements

SPECIFICATIONS (Cont'n)

Item	Manufacturer's Specifications ^a	AMTEC Verification ^b
7 Grinding Mechanism		
7.1 Type	ND	Conical Burr
7.2 Dimension, L x W x T, mm	ND	NM
7.3 Means of attachment	ND	Bolted
7.4 Material	ND	Stainless Steel
8 Safety Devices	ND	None
9 Special Features	ND	None

Note: ND - No Data; NM - Not Measured

^a Data obtained from component nameplates

^b Data obtained from actual test results and on-site measurements

RESULTS

Item	Data
1 Conditions of Cacao Nibs	
1.1 Source	Gumaca, Quezon
1.2 Variety	ND
1.3 Dimension, L × W × T, mm	13.2 × 6.2 × 4.1
1.4 Moisture content, wet basis, %	2.0
1.5 Bulk density, kg/m ³	ND
2 Weight of Input, kg	
2.1 Cacao nibs	12.5
2.2 Water	0.5
3 Milling Capacity, kg/h	45.2
4 Operating Time, h	-

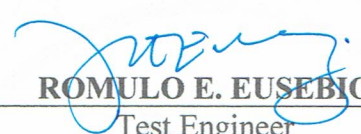
RESULTS (Cont'n)

Item	Data
5 Speed of Components, rpm	
5.1 Motor	
5.1.1 Without load	3595
5.1.2 With load	3529
6 Power Consumption	
6.1 Input to the Controller	
6.1.1 Voltage, V	
6.1.1.1 Without load	219.7
6.1.1.2 With load	213.8
6.1.2 Current, A	
6.1.2.1 Without load	2.5
6.1.2.2 With load	10
6.1.3 Power, kW	
6.1.3.1 Without load	0.55
6.1.3.2 With load	2.14
6.1 Input to the Motor	
6.1.1 Voltage, V	
6.1.1.1 Without load	217.5
6.1.1.2 With load	206.9
6.1.2 Current, A	
6.1.2.1 Without load	4.1
6.1.2.2 With load	8.6
6.1.3 Power, kW	
6.1.3.1 Without load	0.23
6.1.3.2 With load	1.78
7 Noise Level, dB(A)	
7.1 Feeding Operator	
7.1.1 Without load	67.5
7.1.2 With load	83.3
7.2 Bagger	
7.2.1 Without load	72.3
7.2.2 With load	78.8

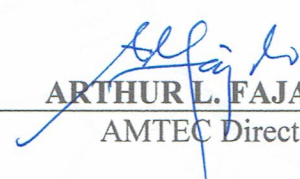
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JUN 13 2019
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